DSC GDP5
Macroeconomics

Web: **https://canvas.uchicago.edu/**

2024

Instructor: Dr. Gina C. Pieters

Course Description

How do we evaluate the consequences of macroeconomic events and policies? This course introduces participants with minimal to no macroeconomic background to macroeconomic policy analysis. We will discuss macroeconomic measurements, how that data relates to our theoretical tools, and why, unlike micro-policy, theory is required for macro-policy analysis. We exchange topic depth for topic range, applying data and theory to analyze economic growth, closed-and-open economy deficit sustainability, monetary policy, fiscal policy, unemployment, inflation, and business cycles.

Prerequisites

Prerequisites: No prior familiarity with macroeconomics is assumed or required. You should be proficient in algebra and interpreting time series graphs.

Course Objectives

After completing the course, students will be able to:

1. Identify components of macroeconomic accounts, how they are calculated, and how they are related.
2. Distinguish agents, markets, and their interactions in a theoretical macroeconomic model.
3. Gather and interpret core macroeconomic time-series data from online repositories.
4. Create an analysis framework that synthesizes data and macroeconomic theory to analyze a policy implementation.

Course Materials

- Data for assignments will come from open data repositories such as GASTAT, IMF, SAMA Source, UN Comtrade, World Bank, and St. Louis FRED.
- Any articles required for reading shall be made accessible via Canvas and added to this syllabus.
- The following textbooks may be helpful if you desire additional reference material for certain topics, but are not required reading
  - Mateer and Coppock: Principles of Macroeconomics
  - Williamson: Macroeconomics
  - Mishkin: The Economics of Money, Banking, and Financial Markets
  - Krugman, Obstfeld, and Melitz: International Economics, Theory and Policy

Course Meetings

Lecture

Lectures will be 3 hours a day (9am to noon) and in-person. Slides will be provided on Canvas prior to class, and may be updated at the end of the day to reflect the material covered. Most classes will be divided into thirds, with time interspersed to work on in-class questions reviewing or applying the current lecture topic. Answers to these in-class activities must be submitted at the end of the day.

Assignments

- **In-Class Exercises (ICE):** These questions will vary significantly in difficulty and scope. Every course meeting will have 3 exercises. You will work in a group of 6-10 people (two to three merged homework groups). These exercises are meant as a *formative* assessment: the intention is to allow learners feedback on their current understanding of a topic in a low-stakes environment. You are expected to share and discuss answers in class. Submit answers *as an ICE group* on Gradescope by 9pm on the due date.
- **Homework Assignments (HW):** There will be four problem sets through course, essentially due on Wednesday and Friday. You must work in groups of 3-5 students. We do not mediate in-group disagreements. Submit answers *as a HW group* on Gradescope by 9pm on the due date.
- **Late submissions** for ICE and HW are accepted for another 3 hours after the due date (until midnight) with no grading penalty, although no assistance will be available after 10pm.
- **Regrade Requests** for ICE and HW must be started within 24 hours after the assignment grades are returned. Regrade requests must be made through Gradescope's "Request Regrade" feature: any discussions during class, TA sessions, or emails are not valid alternatives.

Grades

The grade will be determined by assessments in the following proportions:

- 20% from in-class exercises.
- 20% for each homework assignment. There are four homework assignments.

$$0.2\left(\frac{HW1}{20}\right)+0.2\left(\frac{HW2}{20}\right)+0.2\left(\frac{HW3}{20}\right)+0.2\left(\frac{HW4}{20}\right)+0.2\left(\frac{\text{ICE Score}}{75}\right)$$

The class will be curved to ensure a reasonable grade distribution. 92% or higher will always be sufficient for an A. 70% or higher will always earn at least a C.

Course Policies

Policies on Technical Difficulties and Late Assignments

Given the rapid pace of this course, late assignments are problematic and are not accepted outside the provided three hour window. Groups should structure themselves so that even if one member faces technical difficulties the group's submission is not in jeopardy. It is the responsibility of each group member to make sure their names are on submitted ICE and HW assignments.

Academic Integrity and Honesty

All students are expected to uphold the highest standards of academic integrity and honesty. Ensure your submitted work has genuine attribution as copying is an infringement of academic honesty terms: this means that any work submitted as a group should be that of group members only, without aid of any unapproved outside resources. This means that students shall not represent another's work as their own. The exception to this: You may discuss your answers or findings with other groups *enrolled in the class* to develop your thinking; however, you must give credit to them if you do (unless they are part of your ICE group). Ask if you are uncertain. Each group should download, clean, and compile their own data to gather practical experience in these tasks (and what can go wrong), though the distribution of these tasks across members is left to the group.

Artificial Intelligence (AI) Tools

AI is a part of the workforce now, and students need to learn how to practically and safely incorporate it into their own work. Therefore, students **are** allowed to use any AI powered tool, program, plugin, chatbox or any similar tool to complete assignments, as long as the bot and the prompt used is provided. This applies to any level of AI usage, including providing an answer for AI to rewrite. If you submit an answer created using an AI program, you are certifying that you believe the answer is fully correct. If this answer is incorrect, you will be penalized the same number of points as a student who submitted the same answer without the use of AI.

Schedule

The schedule is tentative and subject to change.

Week 07, 09/30 - 10/04:

- Day 1: Introduction to Macroeconomics and Macroeconomic Data
  - Course Overview // Difference between Micro and Macro Policy Analysis // Lucas Critique // Time Series Terminology and Techniques // Log Transformations of the Data, When and Why. ICE: 9 points.
- Day 2: Introduction to National Accounts and Macroeconomic Measurements
  - GDP and Kuznet's caution // Definitions // Real, Nominal, PPP, and Per Capita // GNI, GNDI, Absorption, Current Account. ICE: 11 points.
- Day 3: The Data and Theory of Long Run Economic Growth
  - Malthusian Growth and Modern Economic Growth: Distinguishing Data Trends // Transition and Steady State Economic Growth // Economic Convergence // Cobb-Douglas Output Function and TFP // Investment and Capital // Growth Accounting // Solow-Swan Model // Growth Policy Analysis: Baseline, Policy, and Interpretation. ICE: 11 points. **HW1 due**
- Day 4: Circular Flow Diagram Representation, Closed Loop Properties, and Implications for Open Economies
  - Agents and Markets in the Macroeconomy // Closed-Economy Circular Flow Diagram // Closed Loop Analysis // Open Economy Circular Flow Diagram // Components of GDP // Trade Policy Naivete // Three Macro-Data Interpretation Checks (real, output share, per-population) // Balance of Payment // Current Account and Financial Account Relationship. ICE: 6 points.
- Day 5: Exchange Rates, International Financial Markets, and External Debt
  - Terminology // Regimes // Law of One Price // Monetary Policy, Interest Rates, and Arbitrage // External Wealth, Debt, and Exchange Rates // Net International Investment Position. ICE: 5 points.

Week 08, 10/07 - 10/11:

- Day 6: Money and Monetary Policy
  - Defining Money // Measuring Money // Transaction Theory of Money // Money Neutrality // Monetary Policy // Real and Nominal Interest Rates. ICE: 7 points. **HW2 due**
- Day 7: Loanable Funds Markets, Intertemporal Consumption, and Fiscal Policy
  - Fiscal Surplus, Deficit, and Debt // Market for Loanable Funds // Equilibrium in Credit Markets // Intertemporal Consumption and Fiscal Policy // Laffer Curve. ICE: 8 points.
- Day 8: Inflation: Measurement and Theory (Long-Run, Short-Run, and the maybe-Phillips Curve)
  - Basket vs. Individual Goods // GDP Deflator vs. CPI // Four CPI Calculations // Price Index Variations // Long Run Inflation // Short Run Inflation // Inflation Expectations // Phillips Curve. ICE: 7 points. **HW3 due**
- Day 9: Unemployment: Measurement and Theory (Labor Market, McCall Search Model, Beveridge Curve)
  - Unemployment "Idea" // Country-Specific Definitions // Household Labor Decisions and Labor Supply // Firm Decisions and Labor Demand // Labor Market Equilibrium // Search and Unemployment // Vacancies and Unemployment // Beveridge Curve // McCall Search Model. ICE: 9 points.
- Day 10: Business Cycles: Dating, Stylized Facts, and Theories
  - Business Cycles // "Stylized Facts" of Business Cycles // Business Cycle Dating and Forecasts // Real Business Cycle Models and New Keynesian Models: Implications for Policy. ICE: 2 points. **HW4 due**